Adrit Panday

GitHub • adrit.panday@mail.utoronto.ca • +1 (437)332-7737 • LinkedIn

EDUCATION

Honors Bachelor of Science, Computer Science Specialist (Co-op) Expected Graduation: October 2027
University of Toronto Scarborough - Toronto, ON

- Minor in Astrophysics and Astronomy
- Relevant Coursework: Operating Systems, Artificial Intelligence, Data Structures & Algorithms, Systems Programming

WORK EXPERIENCE

ML Software Developer, Sunnybrook Hospital || Python October 2024 - May 2025

- Engineered **automated** Python preprocessing **pipelines** for histopathology **tumor** imaging, reducing manual data preparation time by **64%** and enabling faster and more scalable model training workflows.
- Developed a multi-format **image parsing** framework supporting .svs and .tif whole-slide scans with QuPath and OME-XML integration, expanding dataset compatibility across multiple research pipelines by **40%**.
- Optimized **deep learning** training pipelines by implementing custom preprocessing classes for flow recomputation, geometric augmentation, and resolution normalization using **PyTorch**, improving training and evaluation efficiency by **35%**.
- Enhanced AI-assisted tumor segmentation workflows within Paracell's **research** platform while **optimizing GPU**-based preprocessing throughput and memory usage to support large-scale model training on high-resolution whole-slide images.

PROJECTS

GLOW || Node.js, Express, MongoDB, Next.js, Jira May 2025 - August 2025

- Architected and deployed **a full-stack** geospatial data platform enabling **real-time visualization** and analysis of nearshore water temperature datasets across the Great Lakes.
- Containerized application services using **Docker** and implemented CI/CD **automation** via GitHub Actions, ensuring reproducible builds, automated testing covering ~**90%**, and production deployment readiness.
- Designed scalable RESTful APIs and MongoDB data models supporting secure ingestion, retrieval, and manipulation of high-volume geospatial environmental data with JWT-based authentication.
- Engineered an interactive Next.js frontend integrating Leaflet mapping and Chart.js analytics dashboards, enabling dynamic geospatial overlays and improving UI responsiveness by ~**30%**.

Neural Network Training Engine (From Scratch) || C February 2026 - March 2026

- Engineered 1-layer and 2-layer neural networks from scratch in C, implementing forward propagation, full error **backpropagation**, and explicit weight update mechanisms without external ML libraries.
- Trained models on MNIST and CIFAR-10 datasets, achieving high-**80% classification accuracy** on MNIST and analyzing performance gaps between shallow and multi-layer architectures.
- Experimented with hidden layer sizes, activation functions, and input scaling to mitigate saturation and optimize convergence behavior during training.

Multi-Agent Autonomous Navigation Engine (AI) || C January 2026 – February 2026

- Engineered optimal path planning using A* search with admissible Manhattan heuristics and **priority-queue optimization** for real-time navigation in constrained grid environments.
- Developed an adversarial multi-agent decision framework using MiniMax with alpha-beta pruning, reducing evaluated states by ~**60%** at depth ≥10.

AWARDS & ACHIEVEMENTS

F1 Fantasy Optimization, MTA Datathon - 1st Place || Python November 2024 - December 2024

- Identified high performance teams within a set budget by analyzing **15+** data points such as tracks, historic race results and car malfunction data (Secured **1st place** among 50+ participants).
- Applied Python-based trend forecasting, risk evaluation, and budget optimization to maximize score, leveraged historical race data from 2022+ to build predictive models and deliver the winning strategy.
- Collaborated in a 4-person team using agile workflows to divide tasks across data acquisition, model building, and presentation, ensuring efficient execution and clear communication of insights.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, Java
Machine Learning & AI: PyTorch, TensorFlow, NumPy, pandas, Scikit-learn, LLM APIs
Frameworks: Node.js, Express, Next.js
Infrastructure & Tools: Git, GitHub Actions, Docker, Jira, MongoDB, VS Code
Familiar: SQL, Haskell, Assembly, Shell, HTML/CSS